

Nehar Poddar

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Research Profile

Robotics researcher working at the intersection of **reinforcement learning, classical control, and perception**, with a focus on humanoid balance, recovery, and physical interaction.

Research combines learning with structured knowledge of **dynamics, contacts, balance, and 3D environment geometry** to enable humanoids to adapt their behavior to disturbances and their surroundings. Current work focuses on **skill composition, environment-assisted recovery, and perception-to-action learning** for robust physical interaction.

Education

PhD in Robotics – IHMC & University of West Florida Jan 2024 – Present

- Advisor: **Prof. Robert Griffin**
- Research: Humanoid balance and recovery using **reinforcement learning, classical control, and perception**, with emphasis on physics-informed learning and environment-assisted physical interaction.

MSc in Applied Mathematics – Northeastern University Aug 2019 – May 2021

BSc in Mechanical Engineering – NMIMS, Mumbai Jun 2015 – May 2019

Research Experience

PhD Researcher – IHMC / University of West Florida Jan 2024 – Present

- Developed *Bounce Back*, a perception-guided framework that combines **walking, recovery, and environment-assisted interaction**, allowing a humanoid to use nearby walls and tables as physical supports.
- Built a **perception-conditioned mixture-of-experts policy** by combining walking, recovery, and bracing specialists through a learned gate conditioned on **3D occupancy and proprioception**.
- Trained the unified policy with **PPO and advantage-weighted imitation**, using frozen specialists as behavioral targets while learning residual corrections for adaptation.
- Designed a **physics-informed asymmetric actor–critic** using center-of-mass dynamics, capture point, and centroidal momentum as privileged training signals while retaining deployable observations for the actor.
- Developed a **success-gated curriculum** spanning standing, falling, ankle and hip recovery, stepping, and multi-contact stand-up across body configurations and disturbance magnitudes.
- Achieved **93.4% recovery success across 10,000 trials** on the Unitree H1-2 in simulation, evaluated across MuJoCo and Isaac Lab, five initial configurations, and **0–500 N perturbations**.
- Demonstrated **zero-shot deployment on H1-2 hardware at 50 Hz**, recovering from supine, seated, kneeling, and crouched configurations using proprioceptive sensing.
- Conducted controlled ablations to quantify the contributions of **privileged learning, physics-informed rewards, and curriculum design** to recovery performance.
- Developed **network diagnostics and training instrumentation** to analyze per-layer weights, gradients, and observation contributions, identifying unbounded network growth as a cause of intermittent critic divergence.
- Diagnosed **GPU and simulation infrastructure failures** in large-scale Isaac Lab / MuJoCo-Warp training, distinguishing runtime and kernel failures from RL instability using CUDA logs and training diagnostics.
- Tuned **multi-contact simulation and collision solver parameters** for humanoid bracing and recovery, while extending PPO infrastructure with value-loss and critic-regularization methods for stable large-scale training.

Industry Experience

Research Engineer – DEKA Research and Development Corp.

Jan 2021 – Jan 2024

Autonomous Delivery Robot – FedEx ROXO

- Developed **multimodal perception** for an autonomous delivery robot, fusing LiDAR, radar, monocular, and stereo cameras into unified **3D occupancy and drivability representations**.
- Built RGB-D semantic segmentation, stereo depth completion, surface-normal estimation, and 3D reconstruction pipelines for **environment and terrain understanding**.
- Developed **active-learning and uncertainty-estimation pipelines** spanning data collection, model training, evaluation, and deployment on autonomous robotic systems.
- Deployed semantic segmentation for surface classification with **84% performance** and self-attention-based scene-text recognition with **94% performance** on a production robotic platform.
- Implemented **RANSAC, PnP, ICP, and sensor-calibration pipelines in ROS/C++**, reducing calibration time from **60 minutes to 3 minutes**.

Medical Devices

- Modeled insulin dynamics using coupled differential equations and explored **reinforcement learning for adaptive control under partial observability**.
- Developed vision-based flow estimation and temporal signal processing for medical infusion monitoring.

Publications

- **[Work in Progress]** N. Poddar et al., “*Bounce Back: Perception-Guided, Environment-Assisted Humanoid Skill Resumption*,” in preparation, target: *IEEE Robotics and Automation Letters*, 2027.
- N. Poddar et al., “*Embedding Classical Balance Control Principles in Reinforcement Learning for Humanoid Recovery*,” submitted to *IEEE-RAS International Conference on Humanoid Robots*, 2026. [\[arXiv\]](#)
- N. Poddar et al., “*Accelerating Classical Path Planning via Learned Search Space Reduction*,” *AIAA SciTech Forum*, 2026. **Nominated for Best Student Paper.**

Technical Skills

- **Learning & Control:** Reinforcement Learning, PPO, Actor-Critic Methods, Model-Based Control, Optimal Control, Curriculum Learning, Sim-to-Real Learning
- **Humanoid Robotics:** Humanoid Balance, Recovery, Whole-Body Control, Multi-Contact Interaction, Centroidal Dynamics, Capture Point, Contact Modeling
- **Perception & Spatial Understanding:** 3D Occupancy Maps, Depth Estimation, Surface Normals, Semantic Segmentation, Sensor Fusion, LiDAR, Radar, Stereo Vision, RGB-D
- **Robotics Systems:** Isaac Lab, Isaac Sim, MuJoCo, PyTorch, CUDA, ROS/ROS2, Linux
- **GPU Simulation & RL Infrastructure:** CUDA Graph Capture, Newton / MuJoCo-Warp, Contact Solver Tuning, GPU-Parallelized Simulation, PPO Extensions, TensorBoard Diagnostics
- **Programming:** Python, C++, MATLAB, NumPy, OpenCV, Scikit-Learn